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SVG - XII - SET 2

COMMON QUARTERLY EXAMINATION – 2025

S SIVAGANGAI Standard XII

Reg.No.: | | | | | | |

BIOLOGY

Time: 3.00 hrs. **Part - I (Bio-Botany) 35 marks** **Marks: 70**

I. Choose the correct answer. **8 x 1 = 8**

1. Which of the following represent megagametophyte?
   a) ovule b) embryo sac c) nucellus d) endosperm
2. After Fertilization Nucellus transformed into
   a) fruit b) seed c) perisperm d) endosperm
3. Gametes are never hybrid. This is a statement of
   a) Law of dominance b) Law of independent assortment
   c) Law of segregation d) Law of random fertilization
4. Write the ratio for complementary genes
   a) 9 : 7 b) 13 : 3 c) 15 : 1 d) 9 : 3 : 4
5. Changing the codon AGC to AGA represents
   a) missange mutation b) nonsense mutation
   c) frameshift mutation d) deletion mutation
6. An antibiotic gene in a vector usually helps in the selection of
   a) competent cells b) transformed cells
   c) recombinant cells d) none of the above
7. The prevention of large scale loss of biological interity
   a) biopatent b) bioethics c) biosafety d) bio fuel
8. In sugarcane, number of chromosomes (2n) are
   a) 10 b) 34 c) 16 d) 80

II. Answer any 4 questions. **4 x 2 = 8**

9. Write short note on Poland kitt.
10. Draw and label the parts of mature microscope (or) male gametophyte.
11. What are multiple alleles?
12. Define Gene mapping.
13. Name the chemicals used in Gene transfer.
14. What is explant?

III. Answer any 3 questions. (Q.No.19 is compulsory) **3 x 3 = 9**

15. Give the names of the scientists who rediscovered Mendalism.
16. Write about the differences between linkage and crossing over.
17. Write the advantages of Bt-cotton.
18. Name the various nutrient medium used in tissue culture.
19. Draw and label the parts of mature Embryo sac.

IV. Answer all the questions. **2 x 5 = 10**

20. a) Write the suitable diagram explain the structure on an ovule.

(OR)

b) Differentiate continuous variation with discontinuous variation.

21. a) Draw the diagram of different types of aneupbidy with explain.

(OR)

b) What you know about the word PBR_{322}.

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Part - II (Bio-Zoology) 35 marks

I. Choose the correct answer. 8 x 1 = 8

1. The Androgen Binding Protein (ABP) is produced by
   a) leydig cells b) hypothalamus c) sertoli cells d) pituitary cells
2. Placental tissue is tested for chromosomal abnormalities in
   a) ultrasound scanning b) chorionic villus sampling (CVS)
   c) foetoscope d) ZIFT
3. Patau's syndrome is also referred to as
   a) 13 - Trisomy b) 18 - Trisomy c) 21 - Trisomy d) None of these
4. Meselson and Stahl's experiment proved
   a) transduction b) transformation c) DNA is the genetic material
   d) semi conservative nature of DNA replication
5. The theory of recapitulation or biogenetic law was proposed by
   a) Heinrich b) Wallace c) Ernst Von Haeckel d) August Weismann
6. Amphetamines of stimulants of the CNS, whereas barbiturates are
   a) CNS stimulant b) hallucinogen c) CNS depressant d) (a) & (b)
7. Assertion (A) : Plasmodium vivax is a digenic parasite
   Reason (R) : Primary host of plasmodium vivax is man
   a) A and R are true, R is the correct explanation of A
   b) A and R are true, R is not the correct explanation of A
   c) A is true, R is false d) Both A and R are false
8. The most common substrate used in distillers for the production of ethanol is
   a) soya meal b) ground gram c) molasses d) corn meal

II. Answer any 4 questions. 4 x 2 = 8

9. What is Inhibin? State its function.
10. What is criss-cross inheritance?
11. What is meant by ribozyme?
12. Write notes on connecting link with suitable examples.
13. What are the four types of Dendritic cells?
14. Name the bioactive molecules used as an immunosuppressant. Name the microbes that produces this molecule.

III. Answer any 3 questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. Which type of reproduction is effective? Sexual or Asexual?
16. What is amino centesis? Why a statutory ban is imposed on this technique?
17. When does antibiotic resistance develop?
18. State any three goals of HGP.
19. What are the chief applications of antigen-antibody reactions.

IV. Answer all the questions. 2 x 5 = 10

20. a) Describe the structure of human ovum with a neat labelled diagram.

    (OR)

    b) Differentiate between analogous and homologous structure.
21. a) Explain the structure of immunoglobulin with suitable example.

    (OR)

    b) Write about the industrial production of Ethanol. Give its uses.

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COMMON QUATERLY EXAMINATION – 2025 - SIVAGANGAI DIST
HIGHER SECONDARY SECOND YEAR - SCORING KEY

SUBJECT: BIO – BOTANY **MARK: 35**

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Q. NO		ANSWER	MARK
1	B	Embryo sac	1
2	C	Perisperm	1
3	C	Law of segregation	1
4	A	9:7	1
5	A	Missense mutation	1
6	B	Transformed cells	1
7	B	Bioethics	1
8	D	80	1

PART – II

Answer any four of the following **4 x 2 = 8**

9 **Write short note on Pollen kitt.**

1. It is an oily layer forming a thick viscous coating over pollen surface. ½
2. Pollenkitt is yellow or orange coloured and is chiefly made of carotenoids or flavonoids. ½
3. It attracts insects and protects damage from UV radiation. ½
4. Pollenkitt is contributed by the tapetum. ½

2 Mark

10 **Structure of mature microspore**

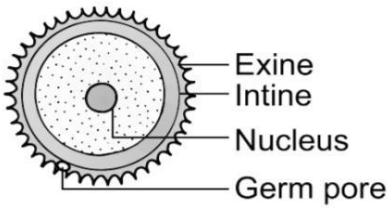


Diagram – 1 Mark
Parts – 1 Mark

2

2 MARK

11 **Multiple alleles**

1. Three or more allelic forms of a gene occupy the same locus in a given pair of homologous chromosomes.

2
2 MARK

12 **Gene mapping:**

1. The diagrammatic representation of position of genes and related distances between the adjacent genes is called genetic mapping

2
2 MARK

13 **Chemicals used in gene transfer.**

1. Polyethylene glycol (PEG) **1**
2. Dextran sulphate. **1**

2 Mark

14 **Explant:**

1. The tissue taken from a selected plant transferred to a culture medium often to establish a new plant.

2
2 MARK

PART – III

Answer any three of the following questions. Q.No 19 is compulsory. **3 x 3 = 9**

15 **Names of the scientists who rediscovered Mendelism.**

1. Hugo de Vries – Holland **1**
2. Carl Correns - Germany **1**
3. Erich von Tschermak - Austria. **1**

3 MARK

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16

linkage - crossing over differences:

linkage	crossing over
The genes present on chromosome Stay close together.	It leads to separation of linked genes
It involves same chromosome of homologous chromosome.	It involves exchange of segments between non-sister chromatids of homologous chromosome.
It reduces new gene combinations	It increases variability by forming new gene Combinations. lead to formation of new Organism

1
1
1
3 MARK

17

Advantages of Bt cotton:

1. Yield of cotton is increased due to effective control of bollworms.
2. Reduction in insecticide use in the cultivation of Bt cotton.
3. Potential reduction in the cost of cultivation.

1
1
1
3 MARK

18

Culture media used for plant tissue culture technique.

1. MS nutrient medium
2. B5 medium
3. White medium
4. Nitsch"s medium.

Any 3
3 x 1 = 3
3 MARK

19

Structure of embryo sac:

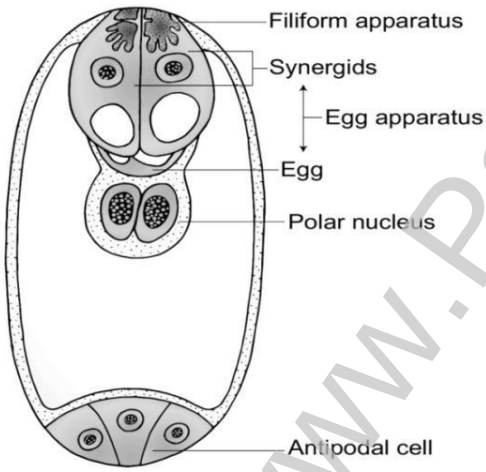


Diagram – 2 Mark + Parts – 1 Mark

3 MARK

PART – IV

Answer all the question **2 x 5 = 10**

20 .A

With a suitable diagram explain the structure of an ovule.

1. Ovule is also called megasporangium.
2. It is protected by one or two covering called integuments.
3. A mature ovule consists of two parts i) Stalk, ii) Body

i) Stalk,

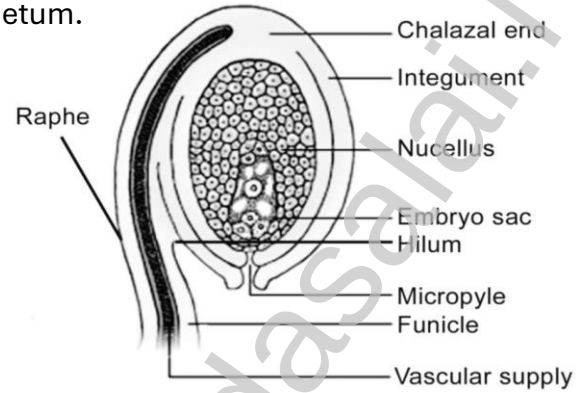
4. The stalk or the funiculus is present at the base and attaches the ovule to the placenta.
5. The point of attachment of funicle to the body of the ovule is known as hilum.
6. The funicle is adnate to the body of the ovule forming a ridge called raphe.

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ii) Body

7. The body of the ovule is made up of a central mass of parenchymatous tissue known as nucellus.
8. It has large reserve food materials.
9. The nucellus is enveloped by one or two protective coverings called integuments.
10. Integument encloses the nucellus except at the top and forms a pore called micropyle.
11. The ovule with one integument said to be unitegmic or two integuments are said to be bitegmic.
12. The nucellus, the integument and the funicle meet at the basal region of the ovule is called as chalaza.
13. The micropylar end of the nucellus contains large oval, sac-like structure called embryo sac or female gametophyte.
14. It develops from the functional megaspore formed within the nucellus.
15. The inner layer of the integument may become specialized to perform the nutritive function for the embryo sac and is called as endothelium or integumentary tapetum.
16. Ex : Asteraceae



Explanation – 3 Marks (Any Six Points) + Diagram – 1 Mark + Parts – 1 Mark

5 Mark

OR

20. B

Continuous Variation - Discontinuous Variation differences:

Continuous Variation	Discontinuous Variation
The phenotype is determined by many genes, and environmental factors.	This variation are controlled by one or two major genes.
This variation due to the combining effects of environmental and genetic factors.	These variations are genetically determined by inheritance factors.
The phenotypic expression is affected by environmental conditions.	The phenotypic expression is unaffected by environmental conditions.
Most of the characteristics exhibit a complete gradation, from one extreme to the other without any break.	Directions of discontinuous variations is unpredictable.
This is also called as quantitative inheritance	This is also called as qualitative inheritance.
Ex : Human height and skin colour	Ex : Style length in primula plant height of garden pea.

Any 5

1

1

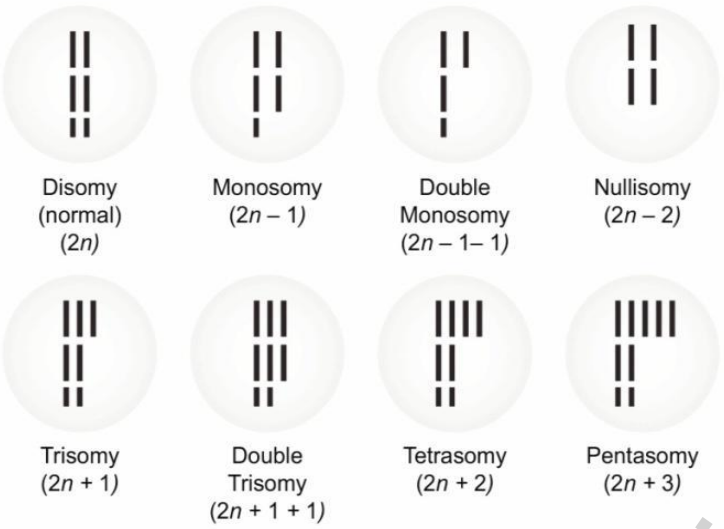
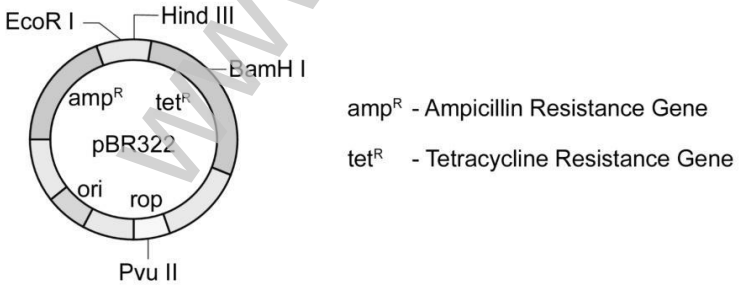
1

1

1

5 MARK

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21. A	**Different types of aneuploidy.** 1. Diploid number is altered either by addition or deletion of one or more chromosomes are called Aneuploidy. 2. They are of two types 1. Hyperploidy 2. Hypoploid  **Explanation – 1 Marks + Diagram – 2Mark + Parts – 2 Mark**	**1** **4** **5 Mark**
	OR	
21. B	**pBR332** 1. pBR 322 plasmid is a reconstructed plasmid. 2. It is most widely used as cloning vector. 3. It contains 4361 base pairs. 4. In pBR, p for plasmid, B and R for scientist Boliver and Rodriguez who developed this plasmid. 5. The number 322 is the number of plasmid developed from their laboratory. 6. It contains ampR and tetR two different antibiotic resistance genes. 7. It is a recognition sites for several restriction enzymes (Hind III, EcoRI, BamH I, Sal I, Pvu II, Pst I, Cla I), ori and antibiotic resistance genes. 8. Rop codes for the proteins involved in the replication of the plasmid.  **Explanation – 3 Marks (Any Six Points) + Diagram – 1 Mark + Parts – 1 Mark**	**3** **1** **1** **5 Mark**

பாரதிராஜா அ
M.Sc., M.Ed., M.A., M.Phil., D.O.A
முதுகலை விலங்கியல் ஆசிரியர்,
தே பிரித்தோ மேல்நிலைப்பள்ளி,
தேவகோட்டை.

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HIGHER SECONDARY SECOND YEAR - SCORING KEY

SUBJECT: BIO – ZOOLOGY **MARK: 35**

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Q. NO	ANSWER	MARK
1	C) Sertoli cells	1
2	B) Chorionic villus sampling	1
3	A) 13 – Trisomy	1
4	D) Semi conservative method of DNA replication	1
5	C) Ernest Van Hackel	1
6	C) Slows down the activity of the brain	1
7	C) Assertion is true, Reason is wrong	1
8	C) Molasses	1

PART – II

Answer any four of the following **4 x 2 = 8**

9	**What is Inhibin? State its functions?** **1.** Sertoli cells are in the testes which secretes a hormone called inhibin. **2.** It is involved in the negative feedback control of sperm production.	**1** **1** **2 Mark**
10	**What is criss – cross inheritance?** **1.** X- Linked traits are inherited from the male parent to his grandson through carrier daughter. This is called criss - cross pattern of inheritance.	**2** **2 MARK**
11	1. Several biochemical reactions in living systems that are catalysed by RNA. 2. This catalytic RNA is known as ribozyme	**1** **1** **2 MARK**
12	**Connecting link** **1.** The organisms which possess the characters of two different groups (transitional stage) are called connecting links. **Example: (Any One)** **2.** *Peripatus* - Connecting link between Annelida and Arthropoda. **3.** *Archeopteryx* - connecting link between Reptiles and Aves.	**1** **2 MARK**
13	Four types of dendritic cells: 1. langerhans, 2. interstitial cells, 3. myeloid and 4. lymphoid cells	**4 x ½ = 2** **2 Mark**
14	**State the importance of cyclosporin A.** 1. immunosuppressant used in organ transplantation is **Cyclosporin A,** 2. it is produced from the fungus *Trichoderma polysporum*.	**1** **1** **2 MARK**

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PART – III

Answer any three of the following questions. Q.No 19 is compulsory. **3 x 3 = 9**

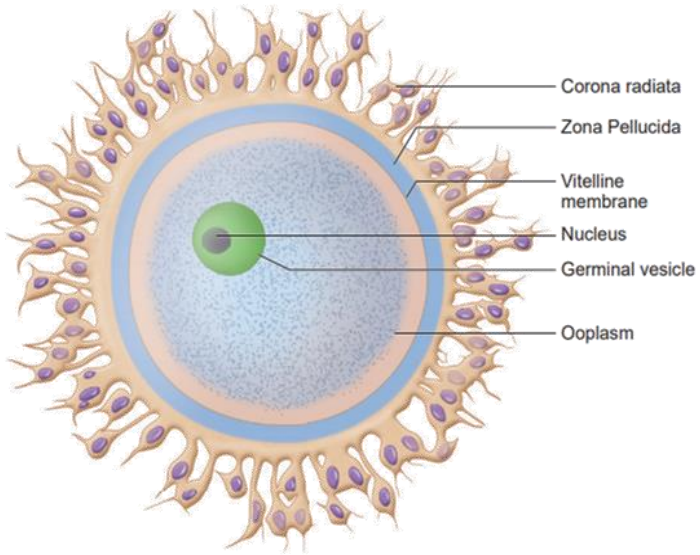
15	**Which type of reproduction is effective - Asexual or sexual and why?** 1. Sexual reproduction is an effective method of reproduction than asexual method. 2. Because, the sexual reproduction introduces variations in organisms. 3. Variations are essential for adaptation and evolution of their own kind.	**1** **1** **1** **3 MARK**
16	**Amniocentesis, the foetal sex determination test, is banned in our country, is it necessary? Comment.** 1. Amniocentesis is a prenatal technique used to detect any chromosomal abnormalities in the foetus. 2. The ban of amniocentesis is necessary in our country. Because, 3. It is misused to detect the sex of the foetus. It creates chance for female foeticide.	**1** **1** **1** **3 MARK**
17	**When does antibiotic resistance develop?** 1. Antibiotic resistance occurs when bacteria develop the ability to defeat the drug designed to kill or inhibit their growth. **Reasons:** 2. Misuse and over use of antibiotics, 3. Poor infection prevention control.	**1** **1** **1** **3 MARK**
18	**State any three goals of the human genome project.** 1. Identify all the genes (approximately 30000) in human DNA. 2. Determine the sequence of the three billion chemical base pairs that makeup the human DNA. 3. To store this information in databases and Improve tools for data analysis. 4. Transfer the related technologies to other sectors such as industries. 5. Address the Ethical, Legal and Social issues (ELSI) that may arise from this project.	**Any 3** **3 x 1 = 3** **3 MARK**
19	**List the application of antigen - antibody reactions.** **1.** To determine blood groups for transfusion. **2.** To study serological ascertainment of exposure to infectious agents. **3.** To develop immunoassays for the quantification of various substances. **4.** To detect the presence or absence of protein in serum. **5.** To determine the characteristics of certain immunodeficiency diseases.	**Any 3** **3 x 1 = 3** **3 MARK**

PART – IV

Answer all the question **5 x 2 = 10**

20 .A	**Describe the structure of the human ovum with a neat labelled diagram.** **1.** Human ovum is microscopic, non - cleidoic and alecithal. **2.** Its cytoplasm is called Ooplasm. **3.** Ooplasm contains large nucleus called germinal vesicle. **4. Outer coat:** Corona radiata, thick with follicular cells. **5. Middle layer:** Zona pellucida, thick. **6. Inner layer:** vitelline membrane, thin and transparent. **7. Perivitelline space:** A narrow space between the vitelline membrane and zona pellucida.	

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5 Mark

Diagram – 2 Mark + Explanation – 3 Mark = 5 Mark

OR

20. B

Homologous structures	Analogous structures.
• Structures which are similar in origin but perform different functions.	• Organisms having different structural patterns but similar functions
• **Ex:** Fore limbs of vertebrates	• **Ex:** the wings of birds and insects
• The thorn in *Bougainvillea* is used as a defence mechanism from grazing animals.	• eyes of the Octopus and of **mammals** and the flippers of Penguins and Dolphins.
• the tendrils of *Curcurbita* used as a support for climbing	• Root modification in sweet potato and stem modification in potato
• It brings about **divergent evolution.**	• It brings about **convergent evolution.**

1
1
1
1
1
5 MARK

21. A

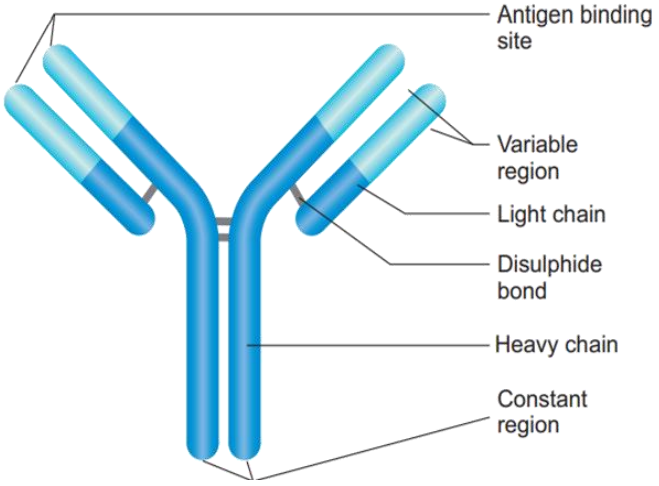
Explain the structure of immunoglobulin with suitable diagram.

1. In 1950s, Porter and Edelman revealed the basic structure of the immunoglobulin.
2. An antibody molecule is Y shaped structure. that comprises of 4 four polypeptide chains.
3. Two identical light chains (L) of molecular weight 25,000 Da.
4. Two identical heavy chains (H) of molecular weight 50,000 Da.
5. The polypeptide chains are linked together by di-sulphide (S-S) bonds.
6. Y shaped structure.
7. Antibody is represented by H2 L2.
8. Each chain (L and H) has two terminals.
9. **Two terminals:** C - terminal (Carboxyl) and amino or N-terminal.
10. **Two regions:** Variable (V) region and Constant (C) region.

5 Mark

Diagram – 2 Mark + Explanation – 3 Mark = 5 Mark

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OR

21. B

The principal substrates:

1. Molasses or corn, potatoes and wood wastes. — **1**

The process of ethanol production:

2. Milling a feed stock
3. Addition of dilute or fungal amylase (enzyme) from *Aspergillus* to break down the starch into fermentable sugars.
4. Yeast is then added to convert the sugars to ethanol. — **2**
5. Then distilled off to obtain ethanol which is upto 96 percent in concentration.

$$\underset{\text{(Glucose)}}{C_6H_{12}O_6} \xrightarrow[\text{Fermentation}]{\text{Yeast}} \underset{\text{(Ethanol)}}{2C_2H_5OH} + 2CO_2\uparrow$$ — **1**

Uses: — **1**

1. Ethanol is often used as a fuel,
2. Biofuel additive for gasoline.

5 Mark

பாரதிராஜா அ
M.Sc., M.Ed., M.A., M.Phil., D.O.A
முதுகலை விலங்கியல் ஆசிரியர்,
தே பிரித்தோ மேல்நிலைப்பள்ளி,
தேவகோட்டை.

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